AMERICAN ASSOCIATION of PHYSICISTS IN MEDICINE

Challenges and Advances in Adaptive Radiotherapy (ART)

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2026 SUMMER SCHOOL
Transforming Technology, Modern Practice, and Clinical Impact
JUNE 16–20 | UNIVERSITY OF MICHIGAN

Disclosures

- Consultant for Varian Medical Systems, Medical Dosimetrist Certification Board (MDCB)

ART – Implementation Gap

- The promise:
  - Address treatment variations
  - Improve target coverage
  - Enhance OAR sparing
  - Superior clinical outcomes and reduced toxicity
  - Standard of care for personalized RT

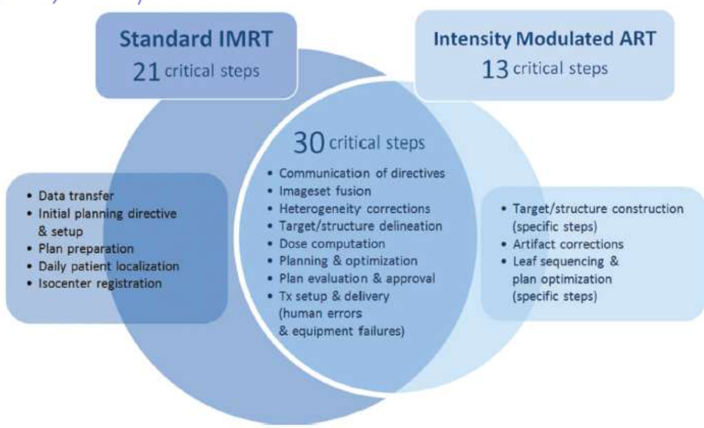
- The reality
  - Complex technologies and workflow
  - Time pressure cascades through every workflow step
  - Increased potential failure points and sources of uncertainties
  - High resource demands and limited scalability
  - Lack of clinical evidence

MEDICAL PHYSICS
The International Journal of Medical Physics Research and Practice

Radiation therapy physics | Free Access

Process-based quality management for clinical implementation of adaptive radiotherapy

Camille E. Noel, Lakshmi Santanam, Parag J. Parikh, Sasa Mutic



- ART Risk priority numbers (RPN) increased by 38% for offline ART
- 75% attributed to **segmentation** and **planning** processes

Contents lists available at ScienceDirect

Physics and Imaging in Radiation Oncology

journal homepage: www.sciencedirect.com/journal/physics-and-imaging-in-radiation-oncology

A practical implementation of risk management for the clinical introduction of online adaptive Magnetic Resonance-guided radiotherapy

FMEA analysis indicated:

- 89 risks were identified with MR guided online ART
- 34% related to MR-specific process and 46% to online ART process
- 66% are high-risk
- **Contouring** and **electron density** are the top two high-risk priority categories

Technical and clinical challenges in ART implementation

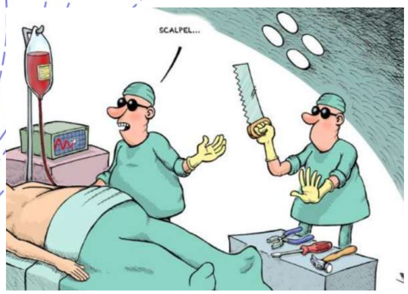
Technical:

- Image quality and registration
- Contouring efficiency and quality
- Electron density accuracy
- Treatment planning quality and robustness
- Dose accumulation
- Patient-specific QA and chart check
- Intrafractional motion

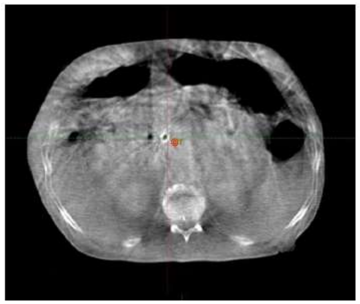
Clinical and logistic:

- Patient selection
- Clinical decision-making for adaptation and plan evaluation
- Multidisciplinary team coordination and workflow management
- Staff training and credentialing
- Cost and staffing

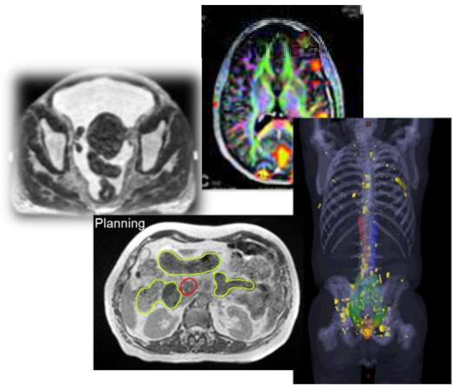
Challenge – image quality for ART



There is no blind surgeon



If we cannot see,
how can we adapt?



- Contouring
- Dose Calculation
- Motion Management
- Clinical Decisions

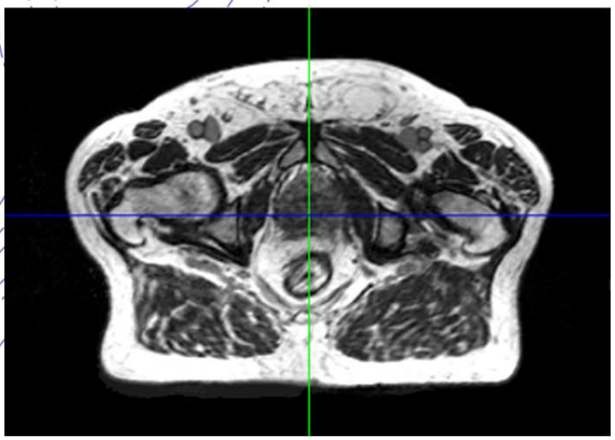
Imaging Technology Innovations

Faster acquisition with higher quality: contrast · geometric accuracy · dosimetry accuracy …

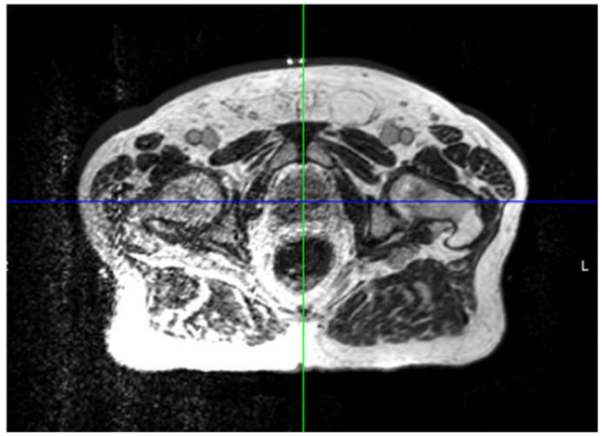
Periodic & Robust IGRT QA

Geometric accuracy & image quality QA of the IGRT system is essential

Simulation MRI

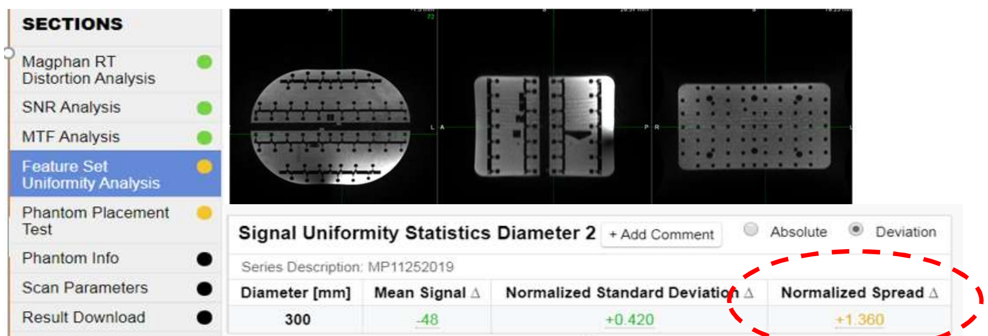


Daily MRI



Multielement RF Coil

Failure in one of the MR coil elements

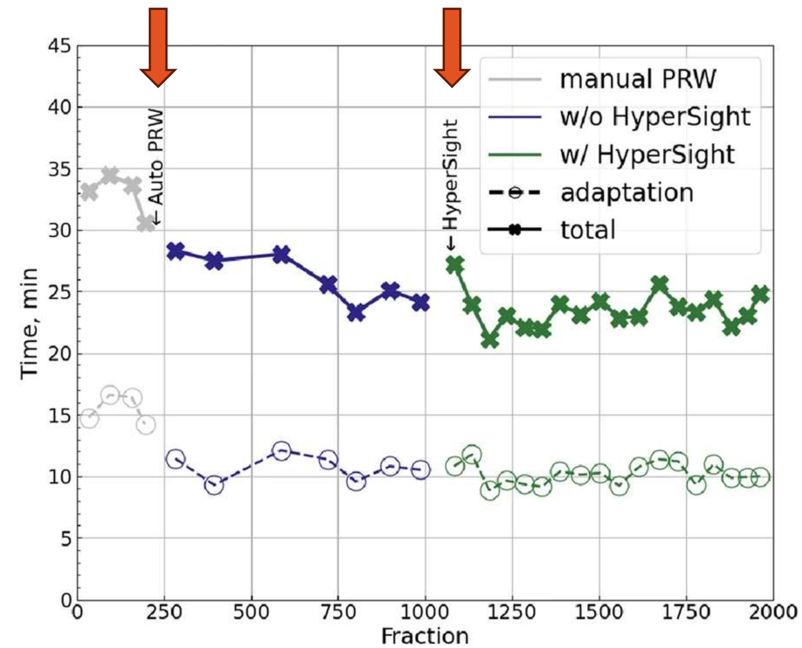


- Uniformity test detected failure in one of the MR coil elements
- Automated phantom image analysis can expedite the process for daily QA

Where dose the clock go?

Contouring Time

Year	Platform	Site	Median (min)	Ref
2017	MRI	Abdomen	10 (5-22)	Lamb Cureus 9(8), 2017
2018	MRI	Abdomen	9 (2-24)	Henke Radiothe Onc. 126 (3) 2018
2021	MRI	Prostate	8 (2-15)	Ugurluer. PRO 11(1), 2021
2025	CT	GYN	10 (7-13)	Sun et al. Front. Oncol. 15(1), 2025
2025	CBCT	Prostate	8 (4-12)	Kalinauskaite et. al. ADRO, 10(10), 2025

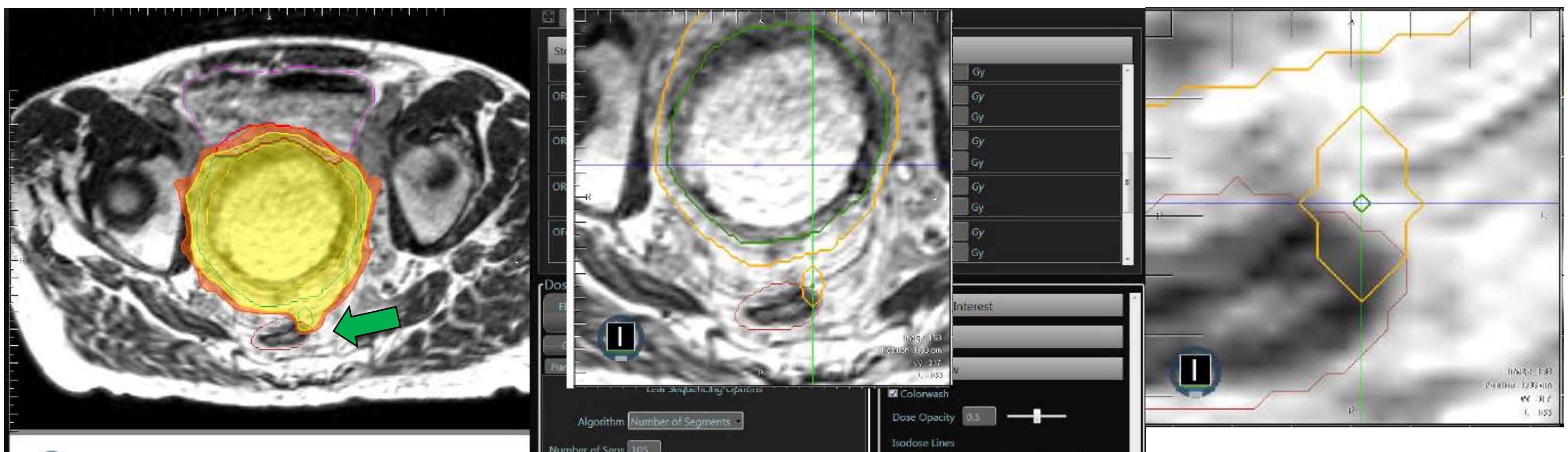


Automated posterior rectal wall contouring reduced Ethos adaptation time from 16.0 → 10.5 min (p<0.001). Workflow optimization + AI = path forward.

Malygina et al. BMC Cancer 2026. 26(1)

Contour quality matters!

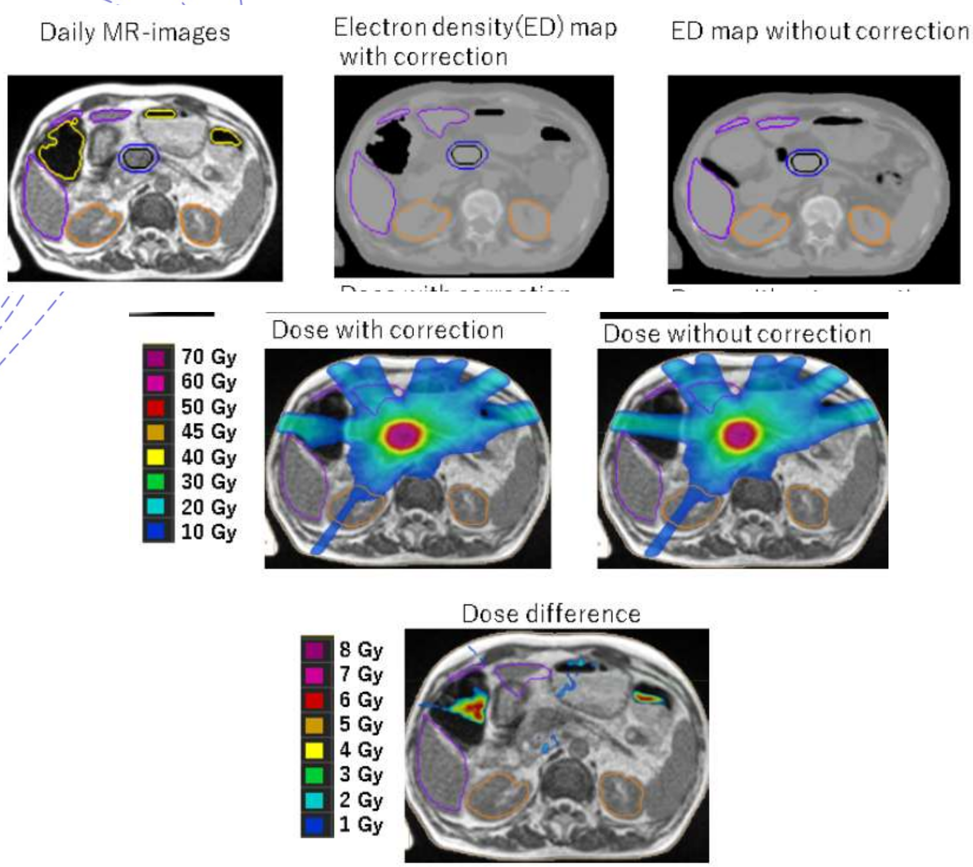
- Inter- and intra-observer variability amplified under time pressure
- Different staff across fractions increases inconsistency
- Critical gaps:
  - Integration of advanced AI-driven autosegmentation into commercial ART platforms
  - Rapid QA methods that maintain accuracy within adaptive time constraints



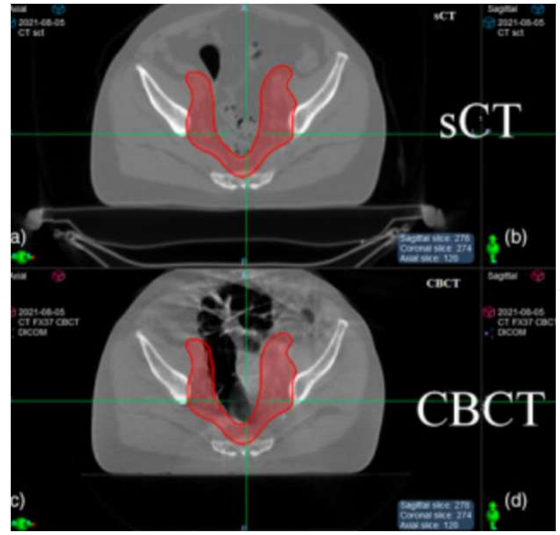
ICRU Report No. 97

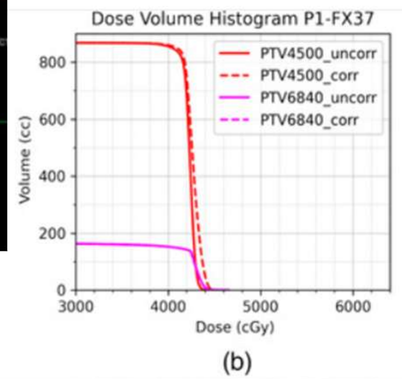
Challenge: Accurate electron density

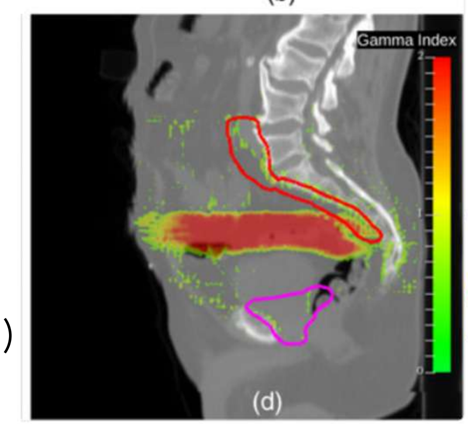
- Accurate electron density is critical for both MR and CBCT guided ART
- To obtain electron density:
  - Bulk assignment
  - Deformable registration with planning CT
  - Synthetic CT
  - Improved HU accuracy with next-generation CBCT
- GI tract air cavities and large tissue inhomogeneities require special handling and correction
- Critical gap: Automated electron density QA and fast detection of ED errors remains an open problem



Chen et al. BJR 2021:94

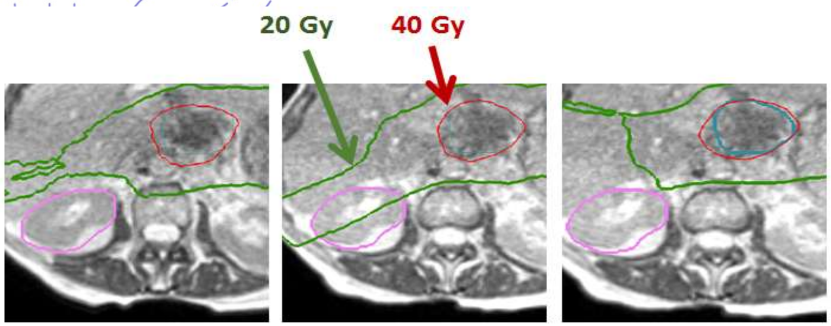




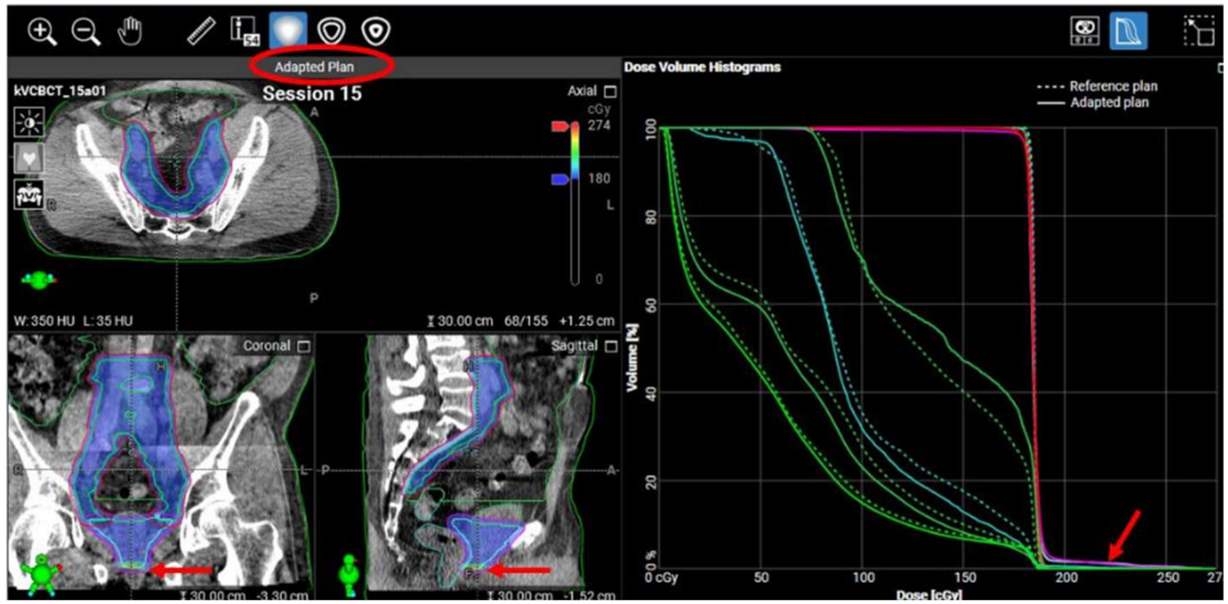


Lems et al. JACMP 2023 24(10)

Challenge: Replanning robustness and efficiency



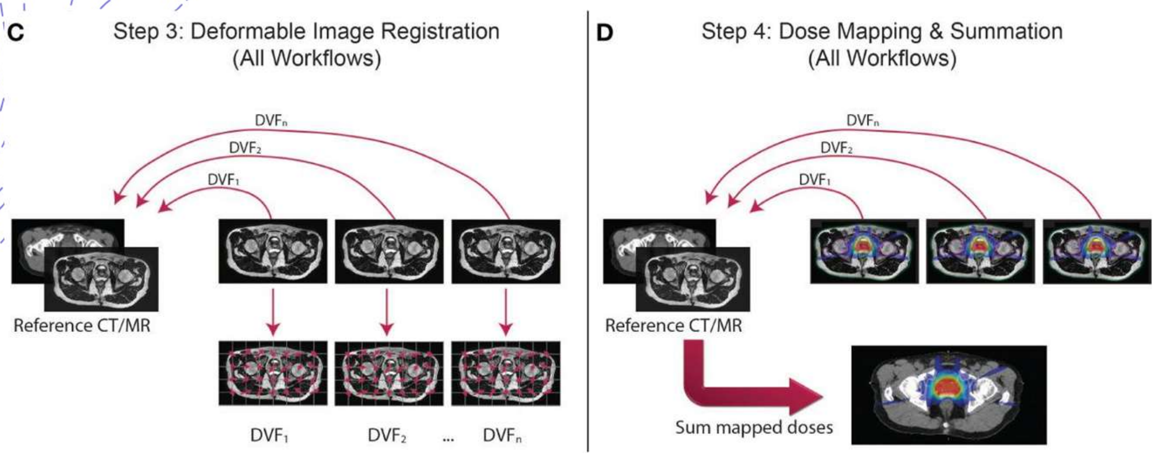
Original plan | 1st adaptive plan | 2nd adaptive plan



- Baseline reference planning enables efficient on-table adaptation within compressed timeframes
- Robust planning to account for potential variations and uncertainties
- Effective QA for plan quality check is critical

Zheng D et al. Cancers 2025, 17, 1462
Lamb J, Cao M et al. Cureus 9(8), 2017

Challenge: Dose accumulation



McDonald BA et al Front Oncol. 2023

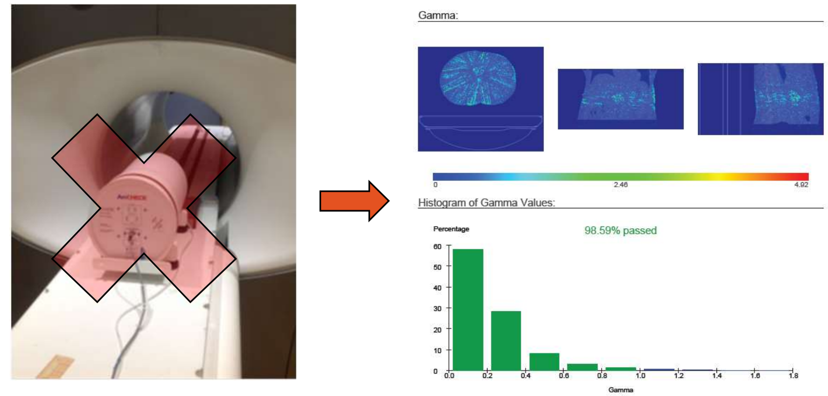
Parotid D_{max} uncertainty up to **33%**; **54%** of voxels with uncertainty **>5%** - *Kos et al. Phys Med 2024*

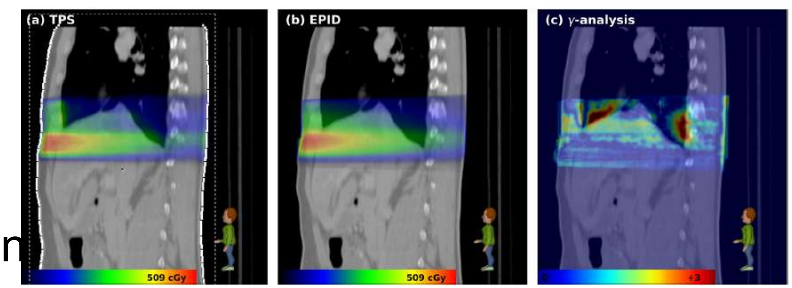
Organs that change shape dramatically between fractions (bowel, rectum) yield **unreliable accumulated dose** – Yoosuf et al. PMC 2025

- DIR – large uncertainties
- Lack of quantitative commissioning and QA methods
- Research-grade tools exist but limited commercial implementation
- Conservative isotoxic approach — each fraction evaluated de novo as if delivered for entire course; limits dose escalation

Challenge – patient specific QA and plan check

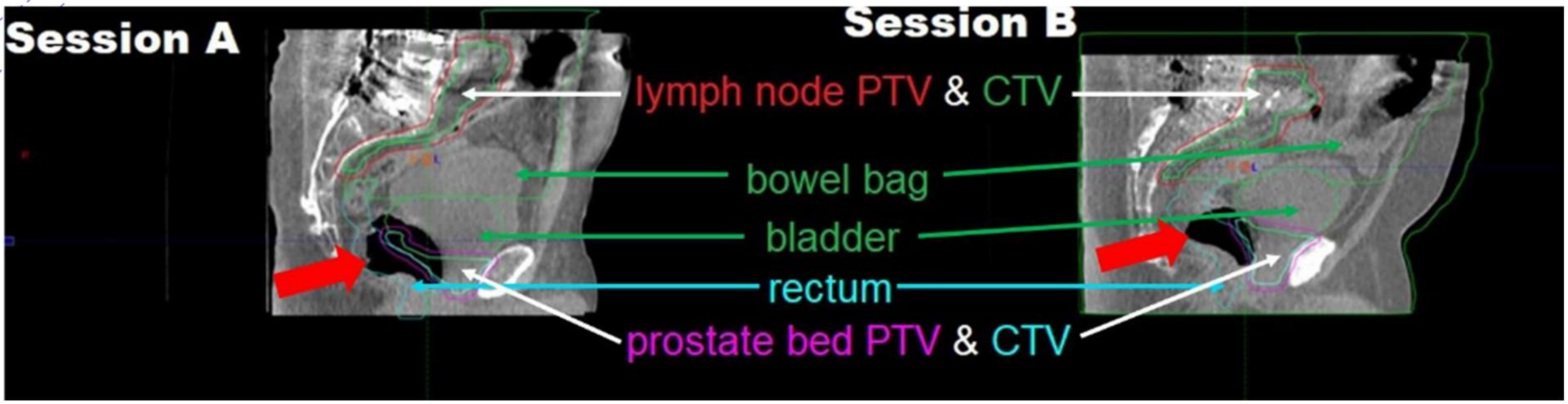
- Traditional measurement-based QA is not feasible for online ART
- Alternative methods:
  - Independent dose calculation
  - Retrospective post-treatment measurement or delivery log analysis
  - “Simulated” treatment delivery without beam-on
  - Real-time monitoring (real-time tracking of machin delivery…)

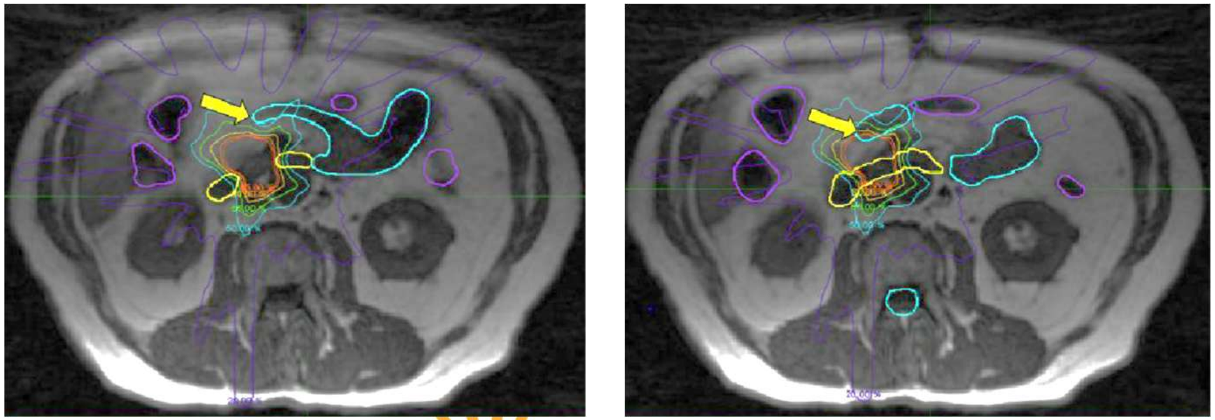




Olaciregui-Ruiz et al. Radiother Oncol 157, 2021

Challenge – Intra-adaptive motion





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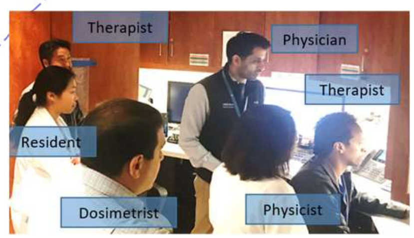
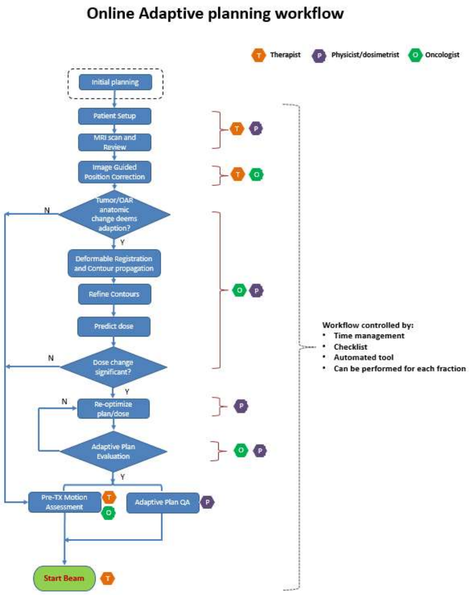
Zheng D et al. Cancers 2025, 17, 1462
Wittmann D, Advances in Radiation Oncology, 2025; 10

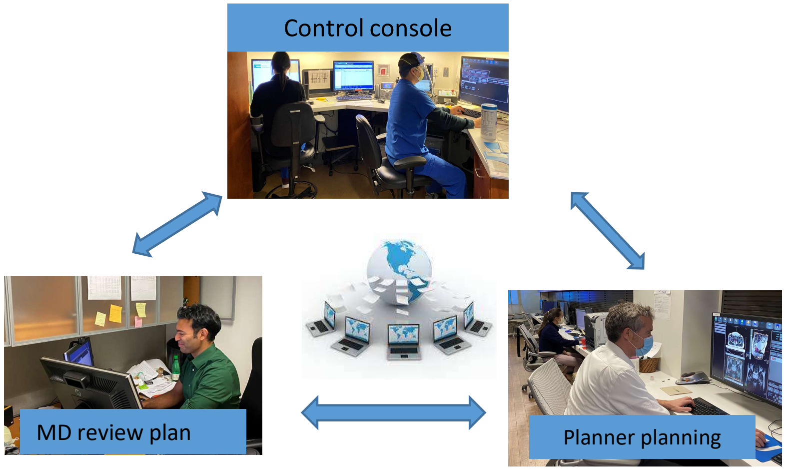
Clinical challenges

- Patient selection:
  - Selection based on disease site and general protocol — lacks individualization
  - Robust predictive model based on patient-specific risk stratification
  - Challenge to predict stochastic variations
- Clinical decision-making criteria and evaluation:
  - No standardized, evidence-based thresholds for when/how to adapt
  - *De novo* decision-making without cumulative dose tracking
  - OAR dose constraints derived from non-adaptive data may not reflect true adaptive tolerances

Workflow/team coordination/training







- Complex workflow and decision-making within compressed time frame
- Traditional RT training inadequate for adaptive complexity
- No standardized competency assessment or credentialing processes

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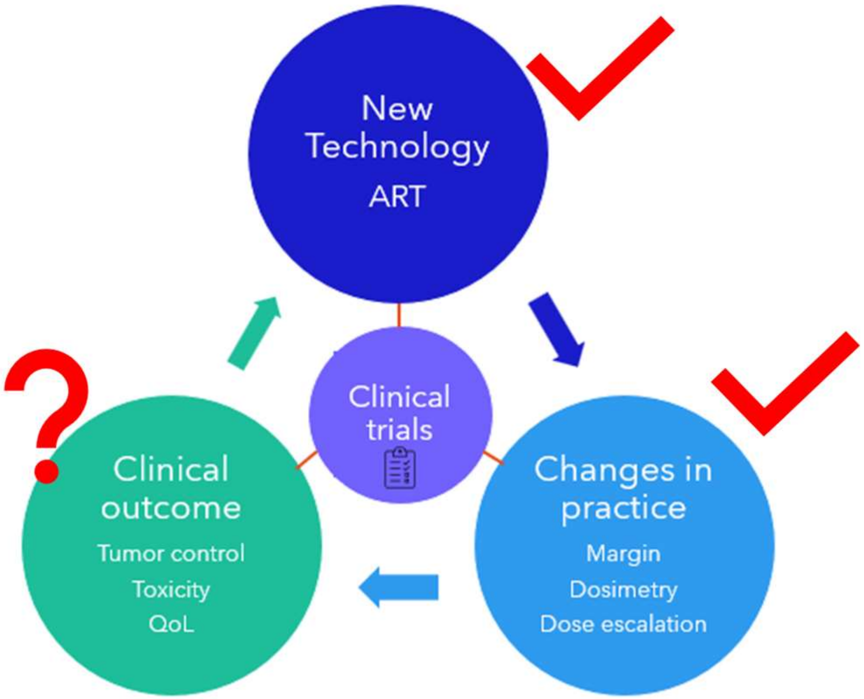
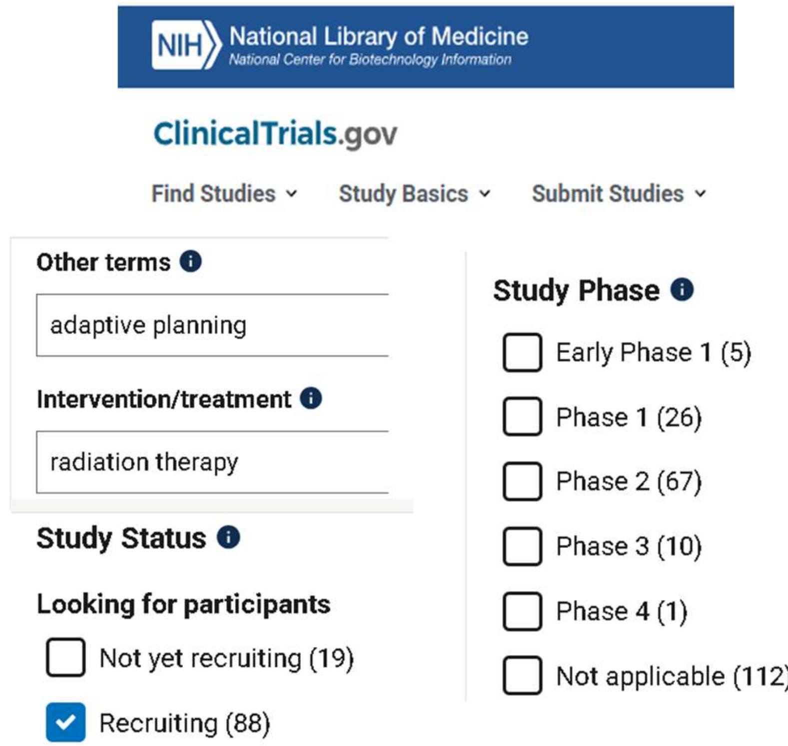
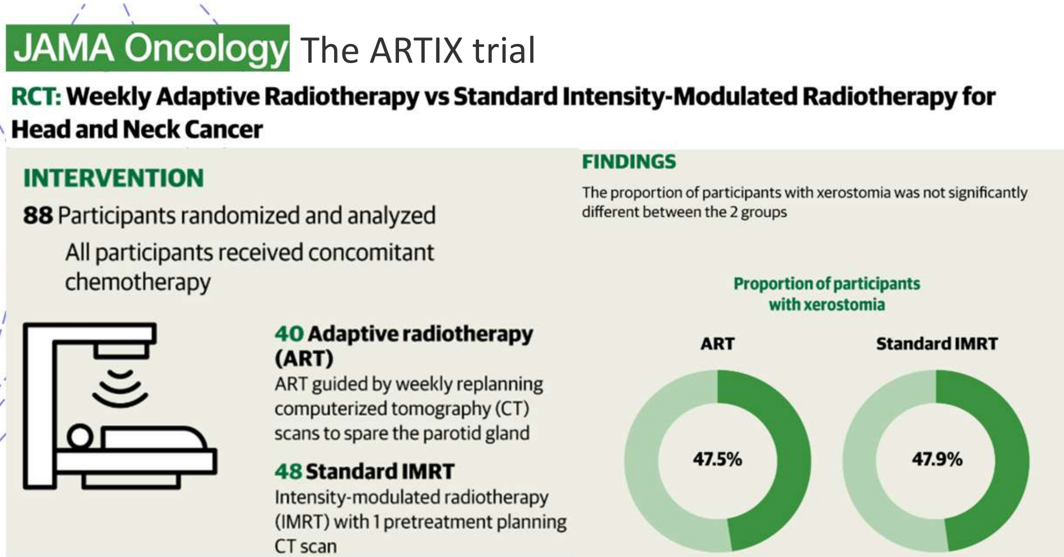
Cost, resource and staffing challenges

- Substantial capital investment: treatment machine + software + site modification + maintenance/upgrade
- Prolonged treatment sessions mean reduced throughput or increased overtime
- Variable ART session durations complicate scheduling and resource planning
- Staffing costs consistently underestimated; no established ART staffing guidelines exist

Clinical evidence is essential





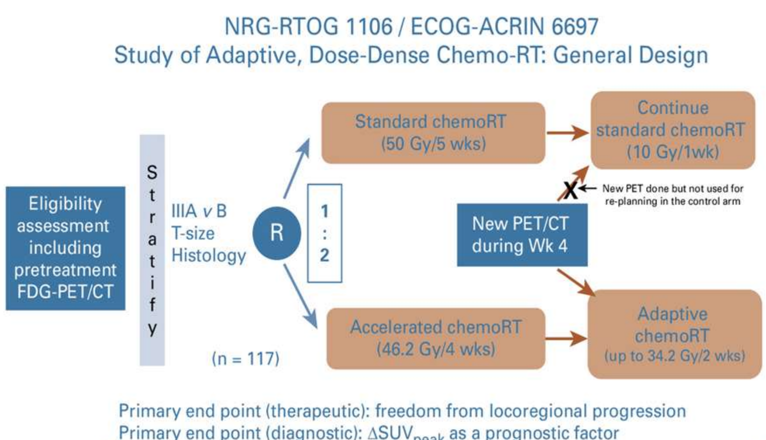


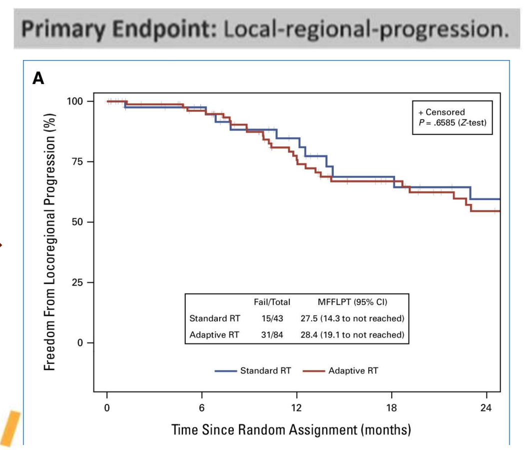
Patient selection

Adaptive timing and frequency

Confounding factors: oral cavity dose, minor salivary glands, plan and setup quality

Margin reduction should be considered





Adapting to a smaller volume has a clear dosimetric benefit

Limitation of single adaptive?

Finding of clinical trial depends on optimal implementation of these technologies

Table 2
Results of QART assessment with patient outcome in prospective clinical trials.

Study [ref]	Type of QA	Number of cases evaluated *n* (%)	Minor deviations *n* (%)	Major deviations *n* (%)	Technical issues with QA review *n* (%)	Impact on clinical outcome	*p* Value
HD 4 [5]	R	368 (98.0)	–	141 (37.5)*	8 (2.1)	7-year RFS with D: 72% *vs.* 7-year RFS with no D: 84%	0.004
EORTC 20884 [2]	R	135 (88.8)	–	63 (46.7)	46 (30.3)	5-year RFS with D: 90% *vs.* 5-year RFS without D: 84%	0.31
RTOG 0411 [4]	R	NS	–	13 (13.4)	NS	Grade GI ⩾ 3 toxicity with D:45%‡ *vs.* Grade GI ⩾ 3 toxicity without D:18%‡	0.05
RTOG 9704 [1]	R	416 (92.2)	–	200 (48.0)**	14/35 (40.0)†	mOS with D: 1.46 yo *vs.* mOS without D: 1.74 yo	0.008
RTOG 0022 [8]	R	67 (97.0)	47 (89.0)	6(11.0)	14/67 (21.0)	LRF with major D: 50% *vs.* LRF with no major D: 6%	0.04
TROG 0202 [15]	P & R††	687 (80.5)‡‡	–	97 (11.8)	33/820 (4.0)	OS with major D: 70% *vs.* OS without major D: 50%	<0.001

Abbreviations: R, retrospective; P, prospective; LRF, local-regional failures; D, deviations; mOS, median overall survival; RFS, relapse-free survival; GI, gastro-intestinal; NS, not specified.

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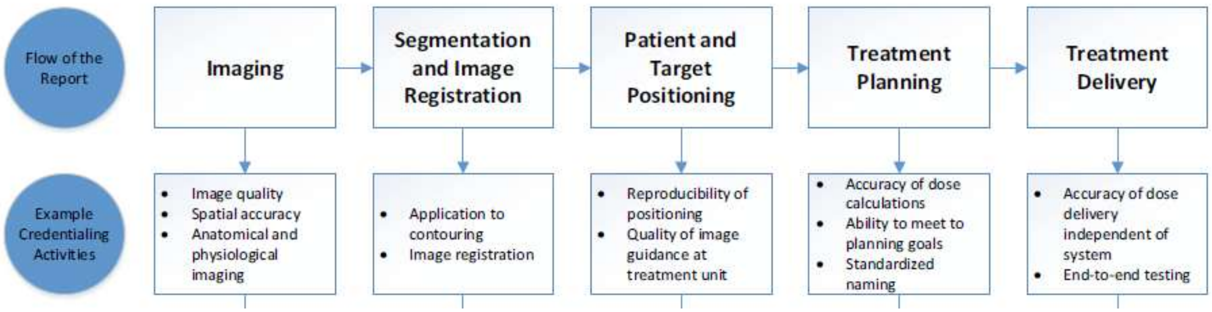
Weber et al. Radiotherapy and Oncology 105 (2012)

Traditional Role – physics support in the execution phase of clinical trial



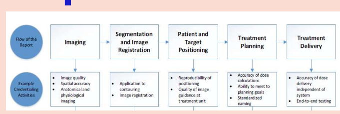
AAPM TG113

Expanded Role – Contributes to all the phases of the technology-driven clinical trial

Trial Design

Technology & protocol review and vetting

Implementation



Analysis & Reporting

Dosimetry/outcome analysis, Failure analysis

Post-trial implementation

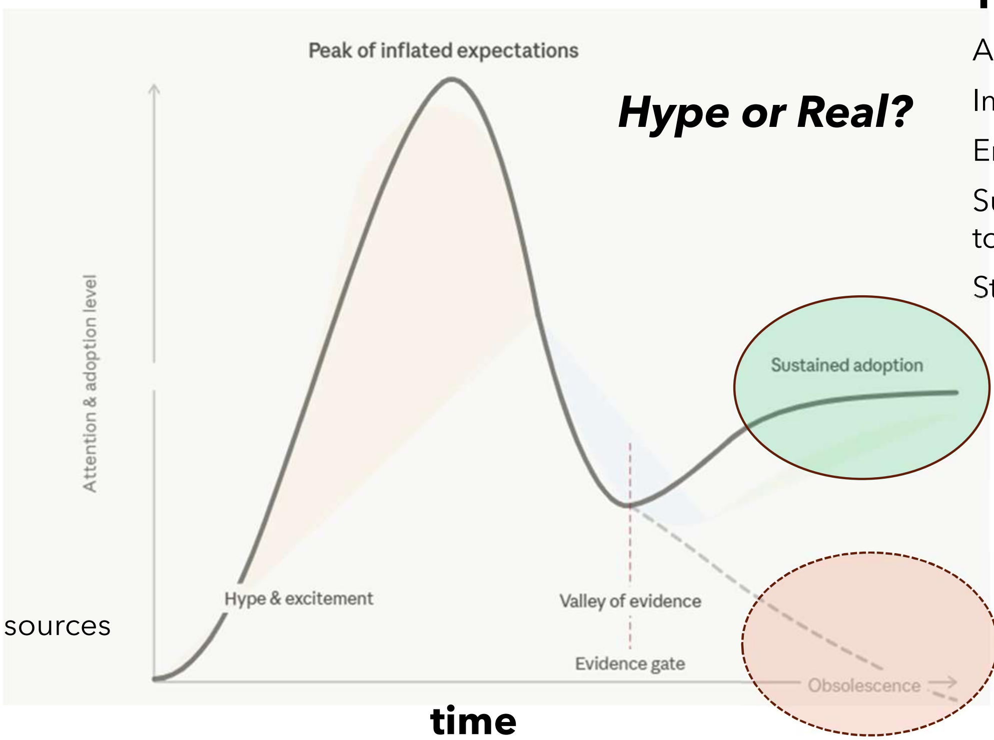
Clinical translation

Technology adoption

Future directions

- ART platform innovations
  - Imaging innovations (speed, quality, multi-modality, functional)
  - RT improvement
- AI-driven ART process
  - Contouring, planning, clinical decision
- Effective QA/QC program
- Clinical evidence: prospective trials and outcome data

Attention & Adoption of Emerging Technologies



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Superior clinical outcomes and reduced toxicity

Standard of care for personalized RT

The challenges

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Thank You!

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